Does a quieter city mean fewer complaints? The sounds of New York City during the COVID-19 lockdown

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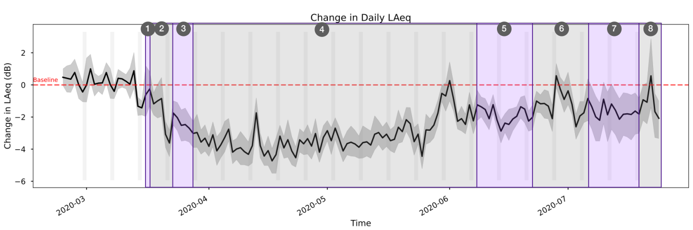
1. Introduction

- As has been widely reported, the world sounded significantly different during the COVID-19 lockdowns in 2020.
- We extend that research to New York City and compare 2020 noise levels, noise class presence, and noise complaints to several prior years to better understand how the urban soundscape and our perception of it changed

2. Dataset

- SPL and audio data from 11 Sounds of New York City (SONYC) sensors active from 2017 to 2020
- Outdoor complaint data from NYC's 311 non-emergency reporting

3. An Unprecedented Drop in Noise



1. Schools closed
2. Bars/restaurants close
3. New York State on pause
4. Construction halt
5. Phase 1 reopening (construction, manufacturing)
6. Phase 2 reopening (outdoor, dining, retail)
7. Phase 3 reopening (indoor dining @ reduced capacity)
8. Phase 4 reopening (low-risk outdoor activities @ reduced capacity)

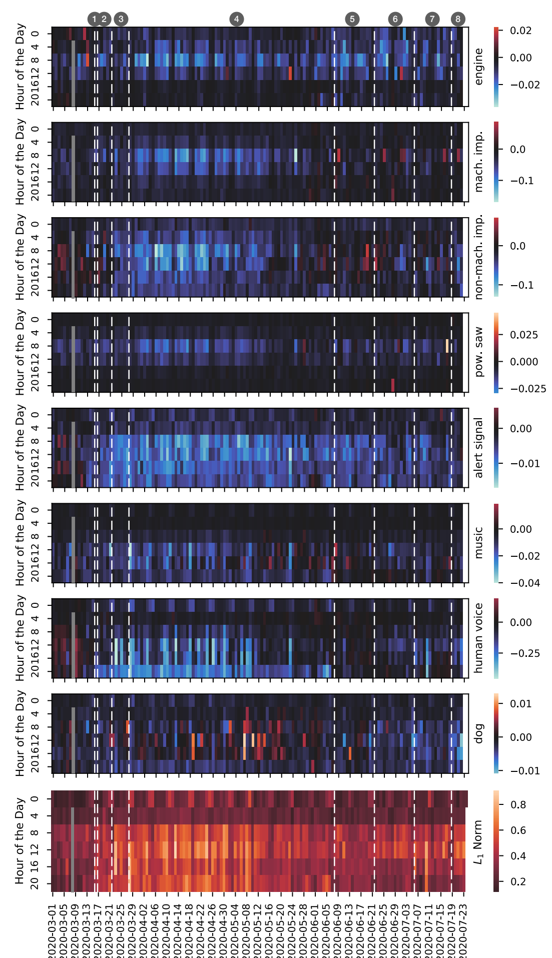
Regression analysis
- Dependent variables: Daily LAeq
- N = 12078
- $R^2 = 0.736$
- '***' indicates $p < 0.001$
- *Seasonal and sensor ID variables not shown for space*

Variable	Coefficient (Std Error)	Standardized Coefficients
City holiday	-1.883*** (0.136)	-0.068
Precipitation (mm)	0.443*** (0.05)	0.042
Temperature (C)	0.078*** (0.006)	0.157
Schools closed	-0.864 (0.695)	-0.006
Bars/restaurants closed	-2.294*** (0.318)	-0.035
New York State on pause	-3.499*** (0.291)	-0.058
Construction halt	-4.147*** (0.094)	-0.231
Phase 1 reopening	-2.624*** (0.198)	-0.066
Phase 2 reopening	-1.758*** (0.193)	-0.044
Phase 3 reopening	-1.443*** (0.21)	-0.034
Phase 4 reopening	-0.942*** (0.32)	-0.014

4. Class-based Change Patterns

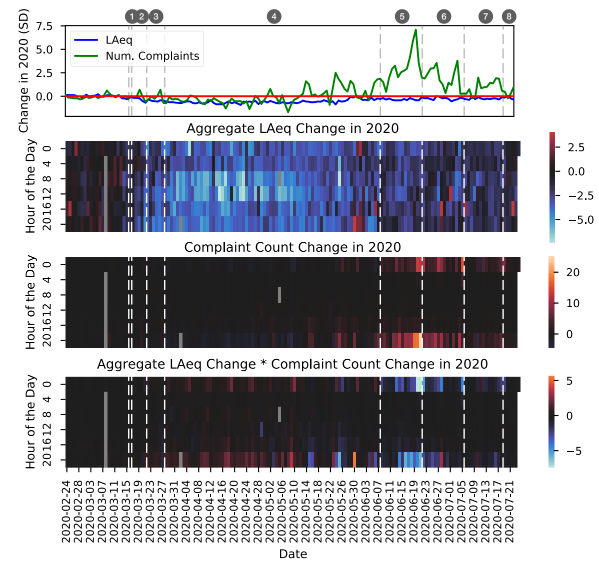
Model:
- Multi-label multi-layer perception w/ 1 hidden layer with 128 ReLU units
- Input: 512-dim OpenL3 features representing for each 1 s frame, 0.5 s hop size
- AutoPool temporal pooling for training on 10 s audio clips
- Output: 8 class likelihoods
- Trained on coarse classes from SONYC Urban Sound Tagging (UST) Dataset v2.3

Output aggregated to class presence ratios for 4-hour intervals



5. A Quieter City Does Not Mean Fewer Complaints

- Change calculated by the difference between 2020 and 2017–2019 data aggregated over sensors



6. Limitations and a Path Forward

- Sound event detection (SED) models predict class likelihood, but that does not quantify impact on noise. We need robust source-specific loudness models.
- Standard fixed vocabulary models detect the expected, but the unexpected is sometimes the most important. We need robust open-set models.
- What insights may we have missed in these unusual circumstances due to these modeling limitations?

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